PROJECT AETHER

Nanotechnology-Enhanced BIPV Glass Architectural Systems

Commercial Pricing & Unit Economics
Market Region: Oman & GCC Region
Currency: Omani Rial (OMR)
Deployment Target: Mid-Rise & High-Rise Facades

AETHER HOME COST	AETHER HOME PRICE	AETHER TOWER COST	AETHER TOWER PRICE
45.00 OMR / m²	**78.00 OMR / m²**	**142.00 OMR / m²**	**245.00 OMR / m²**
Total COGS (Mid-Rise / Commercial)	Commercial Price (42.3% Gross Margin)	Total COGS (Skyscraper Structural)	Facade Price (42.0% Gross Margin)

1. EXECUTIVE SUMMARY & MARKET CONTEXT IN OMAN

Project AETHER integrates active solar energy harvesting into architectural glass using Carbon Quantum Dots (CQDs) and high-efficiency peripheral solar converter strips. Rather than positioning AETHER as a low-cost commodity window, this financial framework models AETHER as a high-performance, active building envelope system priced realistically against GCC construction benchmarks.

Standard high-performance double glazing in Oman costs between **35.00 and 45.00 OMR/m²**, whereas engineered skyscraper curtain wall facades range from **195.00 OMR/m² upward** due to wind pressure engineering (up to 5.0–6.0 kPa), structural anchors, and thermal break profiles. AETHER's commercial pricing strategy combines the local substrate baseline with a proprietary nanomaterial solar layer, maintaining a sustainable **~42% gross margin**.

2. BILL OF MATERIALS (BOM) — AETHER HOME (MID-RISE TIER, 1 M²)

The *AETHER Home* unit is engineered for commercial mid-rise structures and residential builds (up to 8 floors). It features double-glazed 6mm toughened Low-E glass, argon cavity insulation, luminescent CQD coatings, and edge-integrated PV strips.

COMPONENT LAYER	SPECIFICATION & QUANTITY	COST BASE (USD)	COST (OMR)	% TOTAL
Substrate Glass Assembly	6mm Clear Toughened + 6mm Low-E Glass + Argon Gas Cavity	$62.40 / m²	24.00 OMR	53.3%
Carbon Quantum Dots (CQDs)	0.8 g / m² (Green hydrothermal synthesis from citric acid/urea)	$10.40 / g ($8.32/m²)	3.20 OMR	7.1%
Photovoltaic Edge Strips	AlGaAs / Perovskite tandem strips (15% perimeter coverage)	$21.32 / m²	8.20 OMR	18.2%
Encapsulant & Optical Gel	EVA/PVB interlayer film + High-clarity optical waveguide resin	$7.28 / m²	2.80 OMR	6.2%
Framing & Electronics	Thermal break Aluminum frame, internal busbars, micro-inverter wiring	$11.96 / m²	4.60 OMR	10.2%
Direct Assembly Labor	Automated precision coating, edge-sealing, and testing line	$5.72 / m²	2.20 OMR	4.9%
TOTAL DIRECT COGS (AETHER HOME PER 1 m²)			**45.00 OMR**	**100.0%**

3. BILL OF MATERIALS (BOM) — AETHER TOWER (SKYSCRAPER TIER, 1 M²)

Skyscrapers (15+ stories) require specialized structural curtain wall systems capable of resisting extreme wind loads, seismic forces, and high ambient temperatures. The *AETHER Tower* unit utilizes double-laminated 8mm heat-strengthened glass, SentryGlas (SGP) interlayers, and high-density tandem solar strips.

COMPONENT LAYER	SPECIFICATION & QUANTITY	COST BASE (USD)	COST (OMR)	% TOTAL
Structural Substrate Glass	8mm Toughened + 1.52mm SGP + 6mm Low-E Triple Glazed Unit	$273.00 / m²	105.00 OMR	73.9%
Carbon Quantum Dots (CQDs)	1.2 g / m² (High optical density coating)	$10.40 / g ($12.48/m²)	4.80 OMR	3.4%
Photovoltaic Edge Strips	High-efficiency tandem converter strips (20% perimeter coverage)	$31.20 / m²	12.00 OMR	8.5%
Structural Resin & Interlayer	SGP structural interlayers + UV-stable polymer matrix	$13.52 / m²	5.20 OMR	3.7%
Heavy Duty Curtain Framing	Pressure plates, structural gaskets, internal channels & junction box	$28.60 / m²	11.00 OMR	7.7%
Assembly & Quality Control	Pressure testing, optical alignment, and certification testing	$10.40 / m²	4.00 OMR	2.8%
TOTAL DIRECT COGS (AETHER TOWER PER 1 m²)			**142.00 OMR**	**100.0%**

4. SYNTHESIS COSTING & ALGORITHM: CARBON QUANTUM DOTS (CQDS)

Synthesis Cost Algorithm: Green Hydrothermal Conversion of Organic Precursors
Precursor: Citric Acid + Urea | Yield: ~68% Carbon Yield | Bulk Material Cost: 4.00 OMR / kg raw

Commercial scientific suppliers sell laboratory-scale Carbon Quantum Dots at prohibitive rates (250–500 OMR/g). Project AETHER utilizes an in-house hydrothermal green synthesis method using food-grade Citric Acid and Technical Urea.

- **Raw Material Cost:** Citric Acid Monohydrate (0.60 OMR/kg) + Technical Urea (0.35 OMR/kg). Reagent cost per gram of refined CQD = **0.005 OMR/g**.
- **Energy & Thermal Processing:** 200°C hydrothermal reactor cycle consumes ~2.4 kWh per 100g batch. At Omani industrial tariffs (~0.022 OMR/kWh), energy cost = **0.001 OMR/g**.
- **Purification & Dialysis:** Industrial ultrafiltration and freeze-drying cost approx. **1.20 OMR/g** at pilot scale.
- **Total Scaled Cost: 4.00 OMR / gram** ($10.40 USD/g). At 0.8g/m² loading, CQD material costs **3.20 OMR/m²** for AETHER Home and **4.80 OMR/m²** for AETHER Tower.

5. PHOTOVOLTAIC EDGE CELL AREA ALLOCATION

Project AETHER edge-mounts narrow-strip tandem solar cells inside the window perimeter, preserving high visible light transmittance (VLT > 55%) across the main vision pane.

- **Active Cell Area:** 0.15 m² of PV strips per 1.0 m² window unit for AETHER Home (15% coverage area).
- **Cell Material Cost:** High-efficiency tandem cells cost approx. $142.00 USD / m² (54.60 OMR / m² of active cell area).
- **Cost Contribution:** 0.15 m² × 54.60 OMR/m² = **8.20 OMR per 1 m² unit**.

6. COMMERCIAL PRICING STRATEGY & PROFIT MARGIN ANALYSIS

AETHER's commercial price reflects the combined value of high-performance double/triple glazing and active power generation. This structure ensures strong profitability while remaining competitive against expensive imported BIPV modules (280–350 OMR/m²).

METRIC / FINANCIAL PARAMETER	AETHER HOME (MID-RISE)	AETHER TOWER (SKYSCRAPER)
Direct COGS (Cost of Goods Sold)	45.00 OMR / m²	142.00 OMR / m²
Base Glass Market Equivalent (Unpowered)	43.00 OMR / m²	195.00 OMR / m²
AETHER Solar Technology Premium	35.00 OMR / m²	50.00 OMR / m²
Target Wholesale Commercial Price	**78.00 OMR / m²** ($202.80 USD)	**245.00 OMR / m²** ($637.00 USD)
Gross Profit per m²	**33.00 OMR / m²** (42.3% Gross Margin)	**103.00 OMR / m²** (42.0% Gross Margin)
Net Operating Margin (after SG&A / Distribution)	**18.50 OMR / m²** (23.7% Net Margin)	**56.00 OMR / m²** (22.8% Net Margin)

7. VALUE PROPOSITION & RETURN ON INVESTMENT (ROI) FOR DEVELOPERS

Payback Period Algorithm (Muscat Commercial Real Estate):
• Incremental Cost over Base Double Glazing: 78.00 OMR (AETHER Home) − 43.00 OMR (Base Glass) = **35.00 OMR / m² Premium**.
• On-Site Power Generation: ~110 kWh / m² / year under Oman solar irradiation conditions.
• Commercial Electricity Tariff in Oman: ~0.025 OMR / kWh.
• Annual Direct Energy Savings: 110 kWh × 0.025 OMR = **2.75 OMR / m² / year**.
• HVAC Thermal Load Reduction: Low-E + CQD IR absorption reduces cooling load by 15% (saves **2.25 OMR / m² / year**).
• Total Annual Savings per m²: 2.75 OMR + 2.25 OMR = **5.00 OMR / m² / year**.
• **Simple Payback Period: 35.00 OMR / 5.00 OMR annual return = 7.0 Years.**

8. KEY DEFENSE POINTS FOR PITCH PRESENTATIONS & COMPETITION JUDGES

When addressing pitch judges, commercial developers, or technical panels, use the following strategic defenses:

- **Why not sell at OMR 40/m²?** Standard double-glazed glass in Oman already costs 35–45 OMR/m² without any solar capabilities. Selling a complete smart window at 40 OMR/m² would mean operating at a financial loss. Pricing at 78 OMR/m² accounts for both the structural glass substrate and our active nanotechnology layer.
- **Why is AETHER Tower priced at OMR 245/m²?** Skyscraper curtain walls require specialized heavy-duty structural glass engineered to withstand extreme wind pressures (5.0–6.0 kPa). The structural glass alone costs near 195 OMR/m². The 245 OMR/m² price represents only a 25% premium for converting the facade into a clean energy power plant.
- **How do you justify the CQD manufacturing cost?** Lab suppliers sell small quantities at high markups. By producing CQDs in-house via hydrothermal synthesis from cheap agricultural/chemical inputs (citric acid and urea), our raw material cost stays below 0.01 OMR per gram.
- **What is the financial return for building owners?** The 35 OMR/m² technology premium pays for itself in 7 years through direct energy production and reduced air-conditioning loads, on a product designed for a 20-to-25-year service life.